

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization With Tableau Practical – VIII

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Class :- TYIT	Batch :- 2
Date of Assignment :- 27/08/2026	Date/Time of Submission :- 27/08/2026

Performing Advanced Analytics using Trends, Clustering, and Forecasting

A) Add and interpret trend lines (linear, exponential)

This graph shows the relationship between temperature and humidity.

Different colors represent different cities in the dataset.

Trend lines are added to understand the pattern of the data.

It helps us compare how temperature changes with humidity.



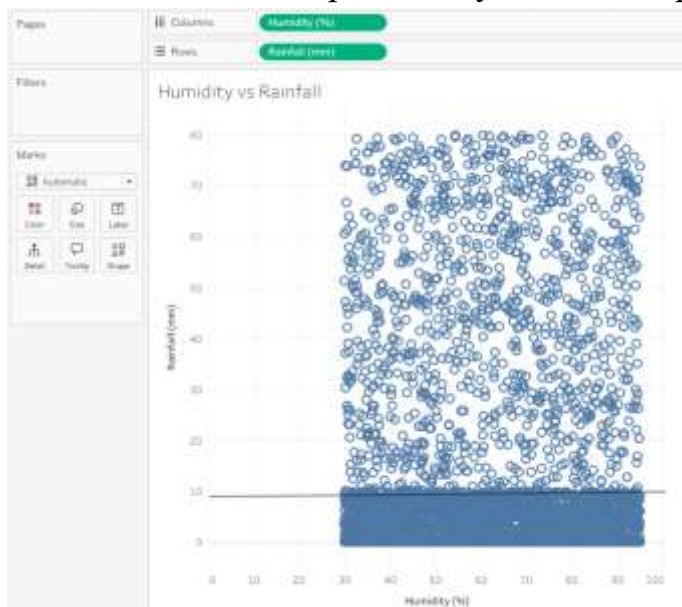
B) Customize and analyze trend models

This graph shows the relationship between humidity and rainfall.

Each circle represents a data value in the dataset.

The points are spread across different humidity levels.

This visualization helps us study the rainfall pattern.



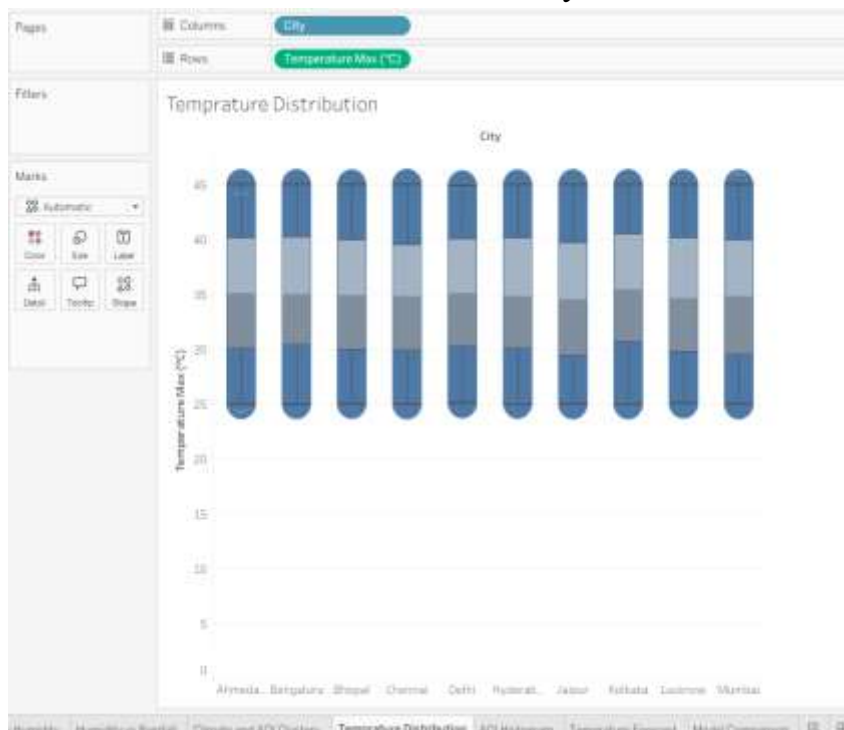
C) Perform clustering on datasets

This graph shows clusters created using climate and AQI data. The data is divided into two groups using different colors. Similar data points are placed together in the same cluster. Clustering helps us identify patterns in the dataset.



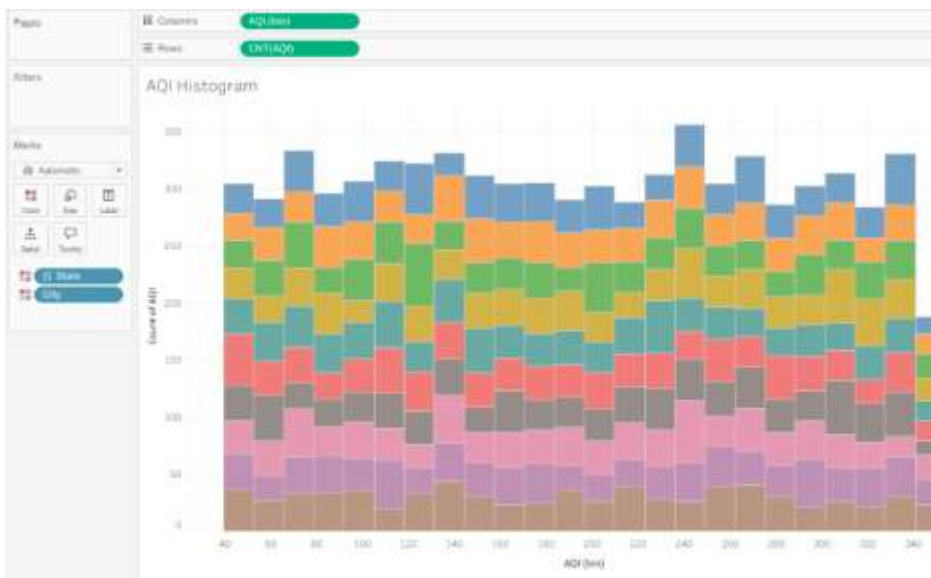
D) Analyze distributions using built-in tools

This chart shows the distribution of temperature for different cities. Each vertical bar represents the temperature range of a city. It helps us compare temperature values between cities. This visualization makes the data easy to understand.



E) Apply forecasting techniques

This histogram shows the distribution of AQI values. The bars represent the number of values in different AQI ranges. Different colors are used to represent different cities. It helps us understand the overall AQI distribution.



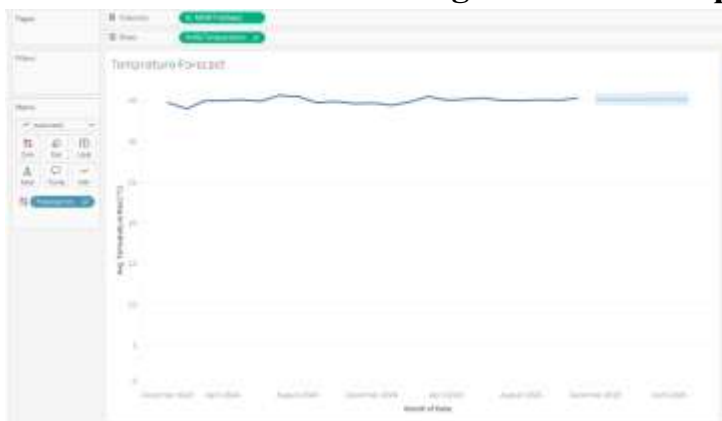
F) Compare different statistical models

This graph shows the forecast of average temperature over time.

The line represents the temperature trend for different months.

Forecasting helps us predict future temperature values.

It is useful for understanding future climate patterns.



G) Export and interpret statistical results

This graph is used to compare different statistical models.

The points show the values of AQI and temperature.

The line helps us understand the overall model trend.

It helps us select and analyze the suitable model.

